DT → Cake from scratch

entropy = $\sum_{i=1}^{n} p_i \log(p_i)$

GINI

→ DT is ok learn → high parameter.

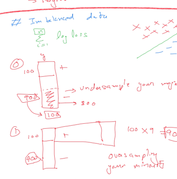
→ Imbalanced data → SMOTE

→ Under fitting ⇔ overfitting

→ Practical problem of memory

→ Regression ...

Imbalanced data



② class weights

$\sum_{i=1}^{n}$ log loss

100 , 900

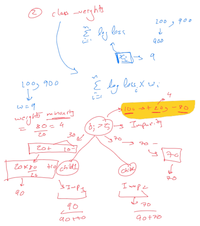
100 , 900

$\sum_{i=1}^{n}$ log loss × w_i

$w_i = \frac{n}{2 \times n_i}$

weight minority

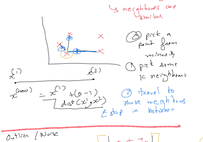
$= \frac{100}{2 \times 100} = 4$



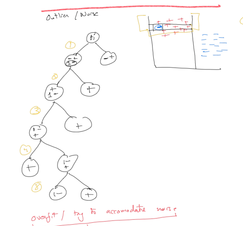
SMOTE (Synthetic Minority Oversampling Technique)

→ oversampling

→ neighbour one hot



Outlier / Noise



overfit / try to accumulate more

↓

DT depth is too too large

→ underfit → high Bias

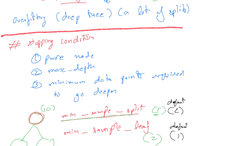
overfitting (deep tree) ⇔ bit of split)

stopping condition

① pure node

② max_depth

③ minimum data points required to get split

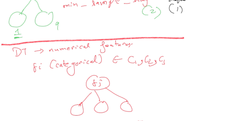


min_sample_leaf

min_sample_split

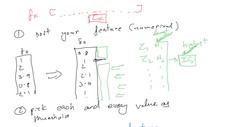
DT → numerical features

f_i: Categorical ∈ C_1, C_2, C_3



for C ..... [Sex]

① sort your feature (numerical)



② pick each and every value as threshold

check → bin num. feature

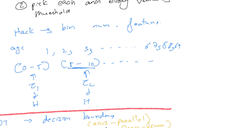
age 1, 2, 3 ..... 62, 63

$C_0 = \{\}$ $C_1 = \{0\}$ ..... $C_{63} = \{0, 1, 2, 3, \ldots, 62\}$

y_i 0 1 1

C_1 1

DT → decision boundary (axis-parallel) (non-linear)



Impact of outliers

KNN → depends → k is very small → Yes

k is large → No

DT → depends → depth is large → Yes

↳ shallow → No

standardize / scale features for DT

No

Tumbleweed (single feature)

↳ entropy / IG

Is your DT → affected by class imbalance

$y = 99\%$ → $G \approx 0$

$y = 1\%$

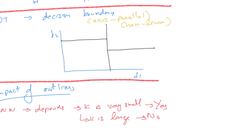
↳ class weights

↳ over sample

↳ under sample

Time complexity

Prediction (DT): O(depth)



Train (DT): $\boxed{n \text{ rows}}$ d (n log n)

n rows

$n \times d \times (n \log n)$

$\approx n \times n \times d \times \log n$

space complexity of DT

O(nodes)

DT → multi-class

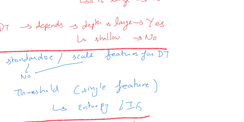
$H = \sum_{i=1}^{n} p_i \log p_i$

$GI = \sum_{i=1}^{n} p_i^2$

→ majority

cat	40
dog	30
house	20

DT → Intro factorable



DT → ◆ regression

Impurity → Leadership ⇔ Target

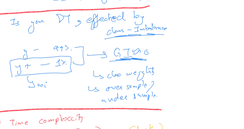
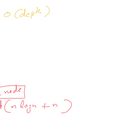
leaf node → predictions

Impurity MSE

$\frac{1}{n_{\text{leaf}}} \sum_{i=1}^{n} (y_i - \bar{y})^2$

↳ $\frac{1}{n'} \sum_{i=1}^{n'} (y_i - \bar{y})^2$

Cabins / Greatness



↳ P (y=1 / x₂)